SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 30marks
Understanding of Concepts	25	24
Experimental Design	30	24
Use of Tools	20	24
Analysis of Results	15	4
Documentation	10	6

Total Marks 22/**30**

Signature of the Faculty

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Engineer to Excel

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Saveetha Institute of Medical And Technical Sciences

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CSA0710 - COMPUTER NETWORKS

TECHNICAL PRESENTATION

Digital Twin Networks and the OSI Model

Guide:

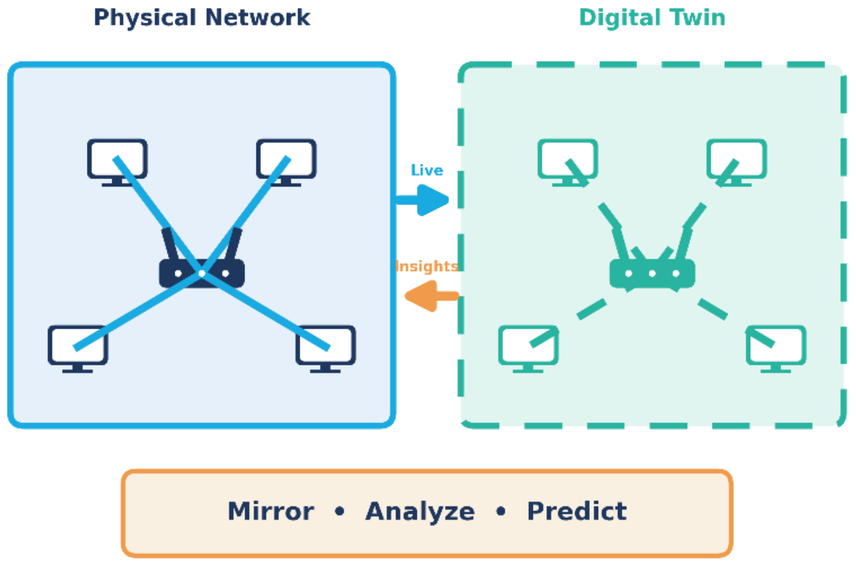
DR.SHANKAR R

Presented by:

J.P.MIRUNALINI JOYS

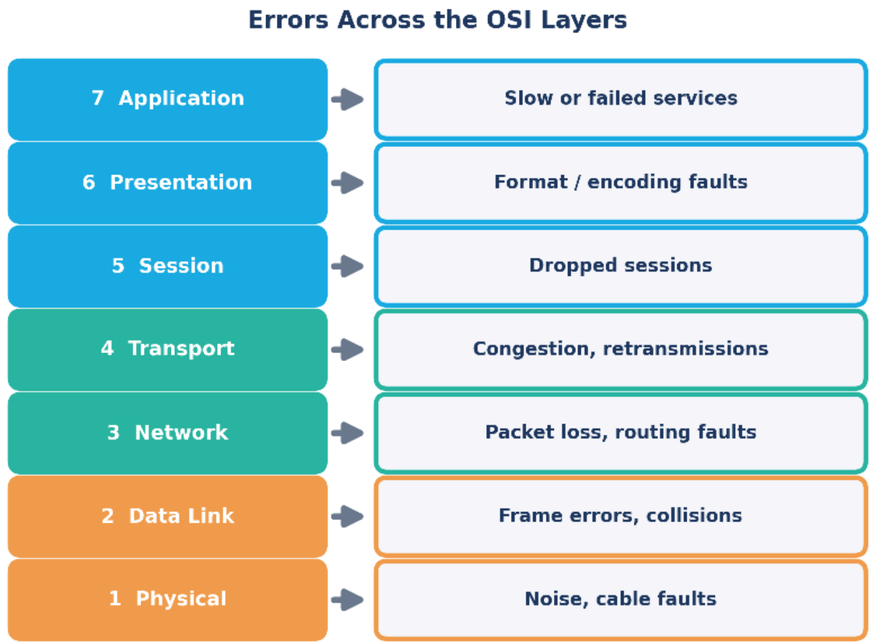
Introduction

- A Digital Twin Network is a virtual replica that mirrors a real network's live state.
- The OSI Model divides network communication into seven layers, each with a clear role.
- Mapping the twin to OSI layers helps locate exactly where a fault occurs.
- Engineers can test changes and failures in the twin without risking the real network.
- Machine Learning on twin data helps detect and predict network errors.



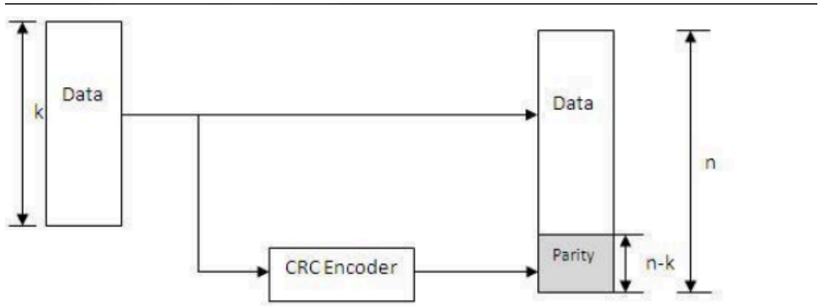
Types of Network Errors

- Physical Layer: Noise, interference, and cable faults corrupt raw bits.
- Data Link Layer: Frame errors and collisions occur between adjacent devices.
- Network Layer: Routing faults and packet loss stop data reaching its destination.
- Transport Layer: Retransmissions and congestion reduce throughput.
- Upper Layers: Application faults and anomalies appear as slow or failed services.

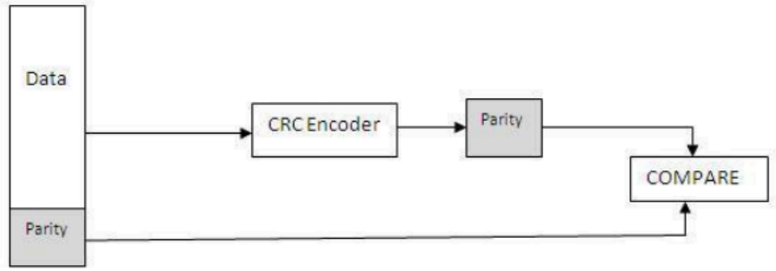


Traditional Error Detection Methods

- Parity Check: One extra bit detects single-bit errors at the Data Link layer.
- Checksum: IP, TCP, and UDP headers use calculated values to detect changed data.
- CRC: The frame check sequence detects burst errors in transmitted frames.
- Rule-Based Monitoring: Fixed thresholds flag problems in the real network.
- Traditional methods cannot learn from twin data or predict future faults.



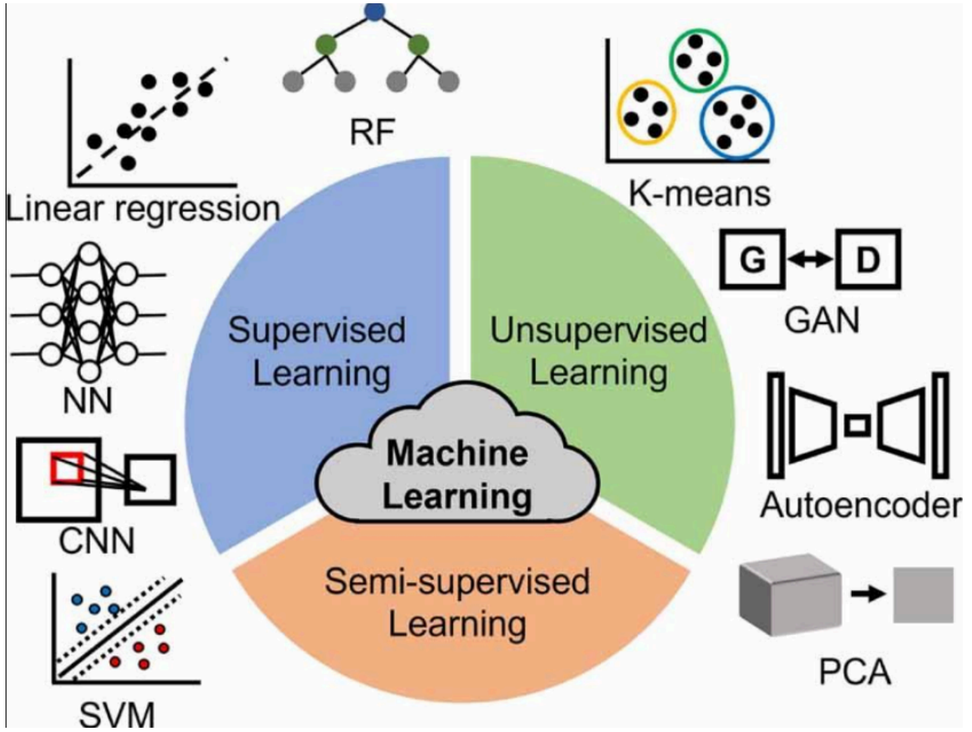
Error detection at transmitter end



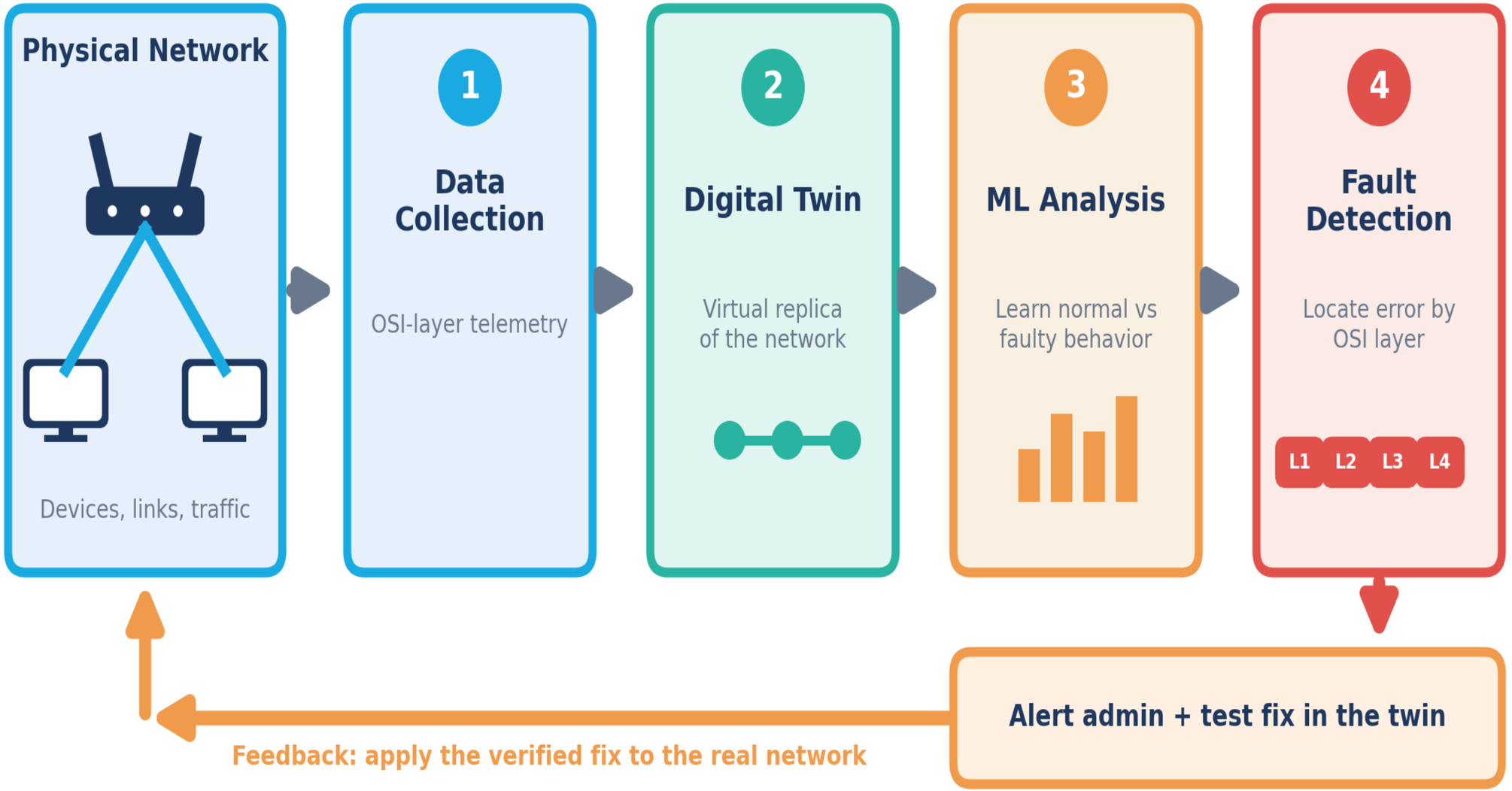
Error detection at receiver end

What Is Machine Learning?

- Machine Learning is a branch of Artificial Intelligence.
- A digital twin keeps supplying live network data for models to learn from.
- Supervised Learning: Uses labeled fault data to classify known errors.
- Unsupervised Learning: Finds unusual twin behavior without labels.
- ML links each anomaly to the OSI layer where it most likely occurs.

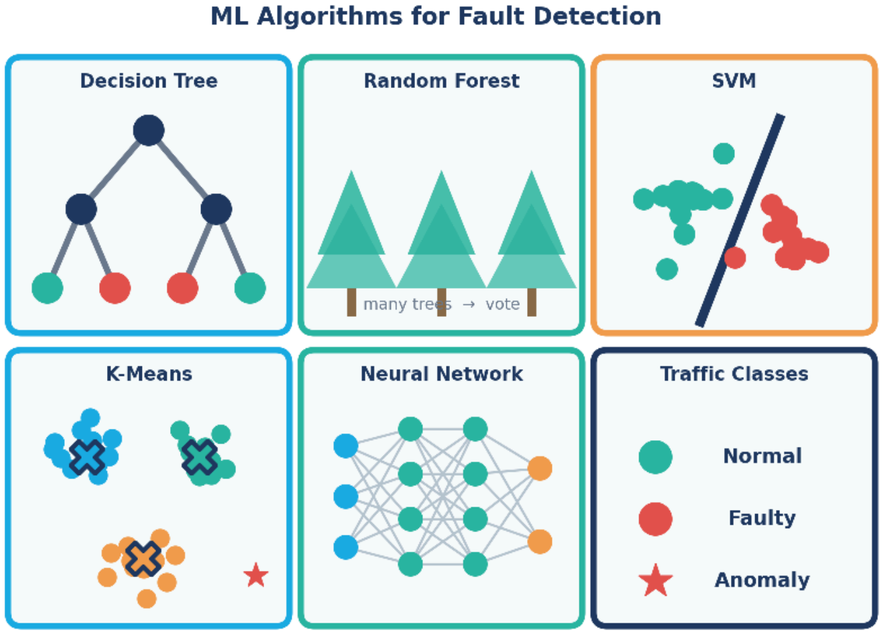


Proposed System Architecture



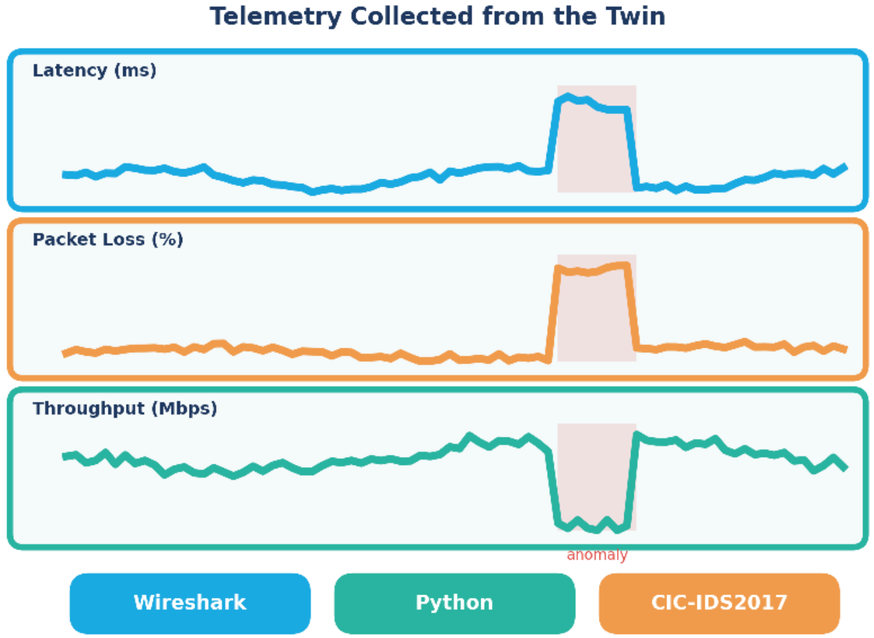
Machine Learning Algorithms

- Decision Tree: Classifies faults using rules over layer-wise metrics.
- Random Forest: Combines many trees for reliable fault classification.
- SVM: Separates normal and faulty traffic classes in the twin.
- K-Means: Groups twin data and helps identify anomalies.
- Neural Networks: Learn complex cross-layer patterns to predict faults.



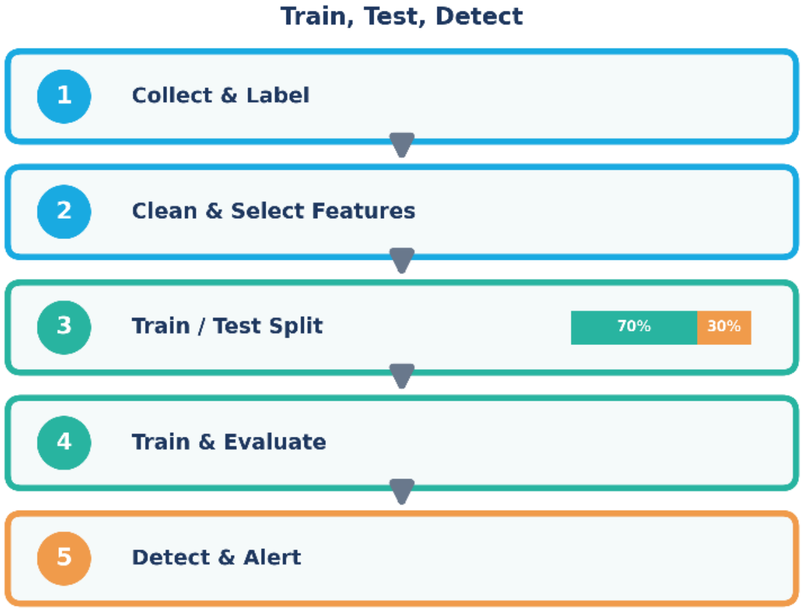
Data Collection and Features

- Collect telemetry from the real network and feed it into the digital twin.
- Measure packet loss and packet error counts at each OSI layer.
- Record latency, jitter, and round-trip time.
- Monitor throughput and TCP retransmissions at the Transport layer.
- Use Wireshark, Python, and datasets such as CIC-IDS2017.



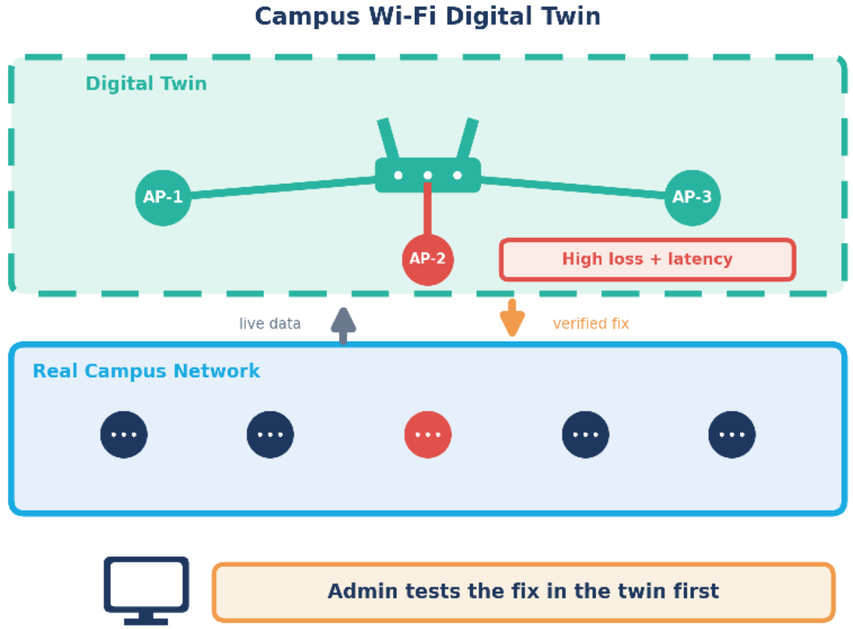
Model Training and Detection

- Collect and label twin data, tagging each sample with its OSI layer.
- Clean the data and select relevant features.
- Split the dataset into training and testing sets.
- Train and evaluate the selected ML algorithm.
- Detect suspicious patterns, generate alerts, and test fixes in the twin.



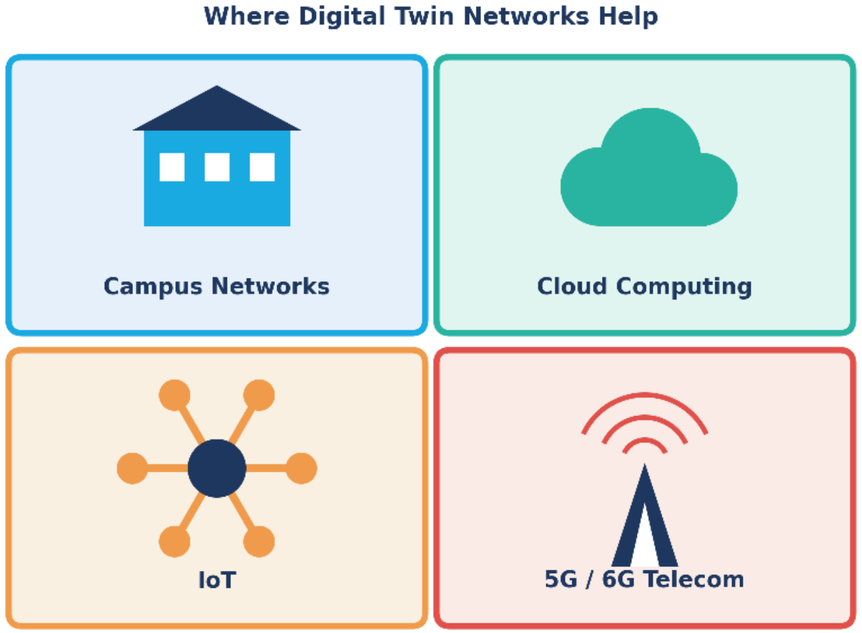
Example Scenario

- A college Wi-Fi network faces frequent connection drops.
- Its digital twin receives live packet loss and latency measurements.
- The ML model analyzes the twin data across the OSI layers.
- It flags a congested access point at the Data Link and Network layers.
- The administrator tests a fix in the twin, then applies it to the real network.



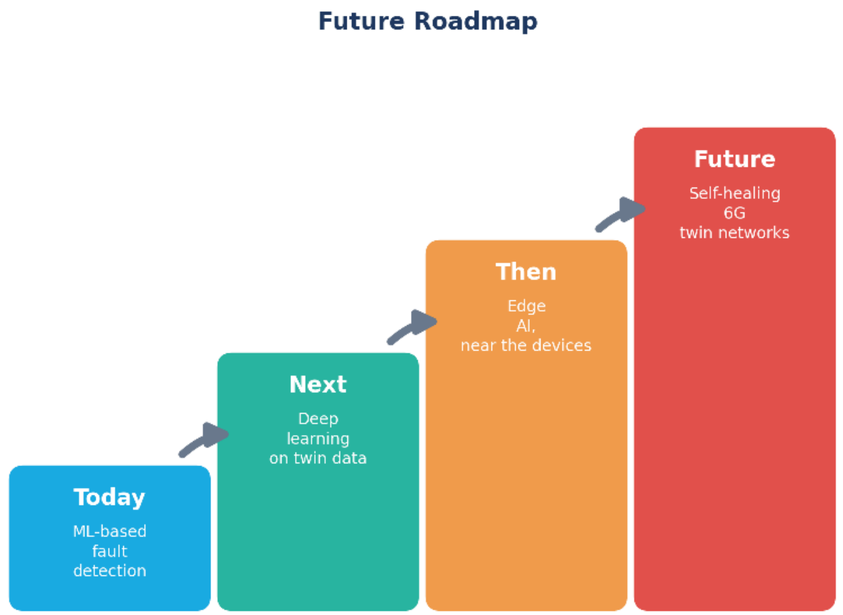
Advantages, Challenges and Applications

- Advantage: Tests changes and faults safely without disturbing the live network.
- Advantage: Layer-wise monitoring makes fault location faster.
- Challenge: Needs accurate real-time data to keep the twin synchronized.
- Challenge: High computing cost and possible false alarms.
- Applications: Campus networks, 5G/6G telecom, cloud computing, and IoT.

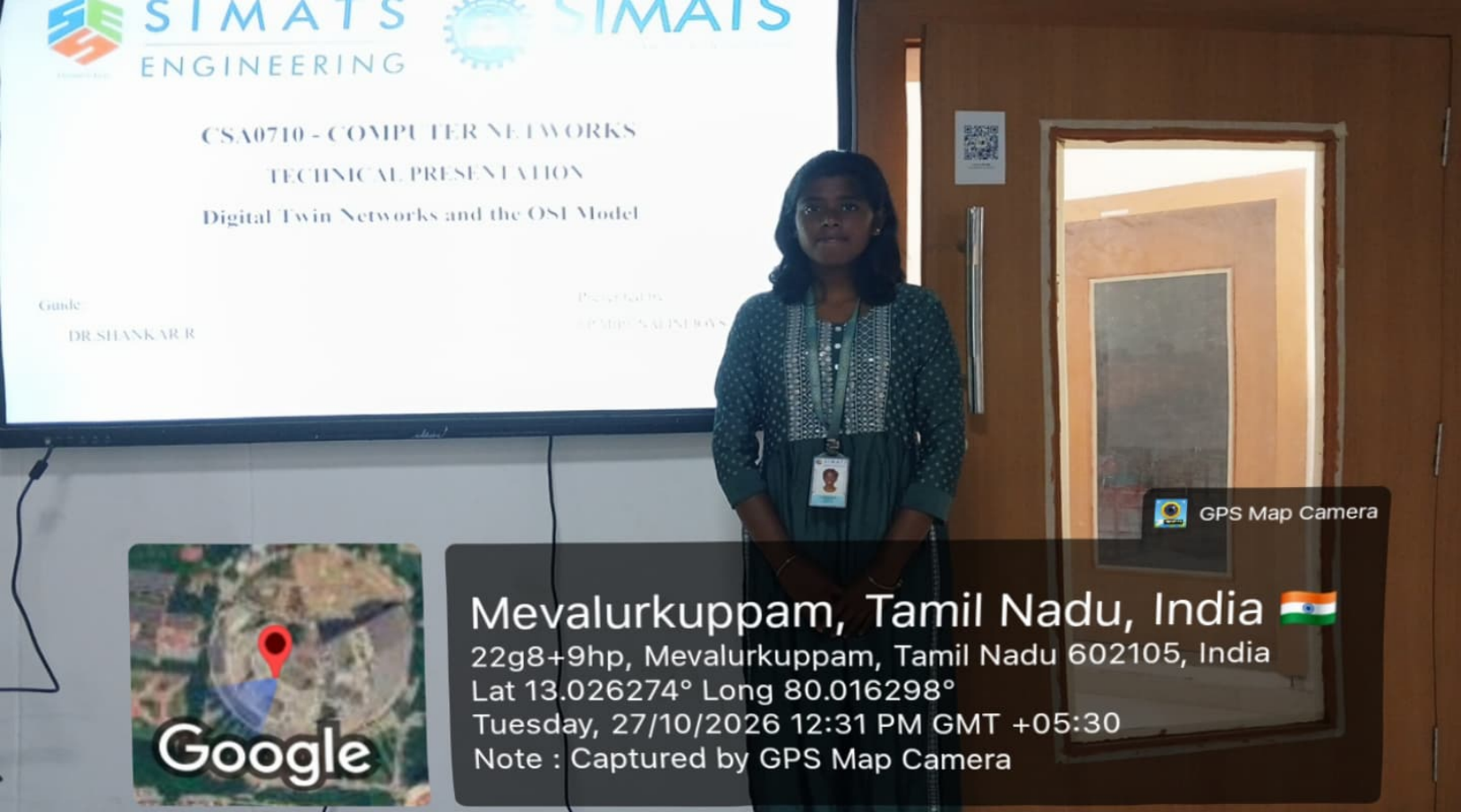


Conclusion and Future Scope

- Digital twin networks give a safe virtual copy for analysis and fault detection.
- The OSI Model helps place each fault at the correct layer.
- ML on twin data complements traditional error detection techniques.
- Future systems can use deep learning, edge AI, and 6G digital twins.
- Self-healing networks can improve reliability and reduce troubleshooting time.




Thank
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TECHNICAL PRESENTATION
Digital Twin Networks and the OSI Model
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Tuesday, 27/10/2026 12:31 PM GMT +05:30
Note : Captured by GPS Map Camera

TECHNICAL PRESENTATION QUESTIONS

1. Role of the OSI Model in Software-Defined Networking (SDN)
2. Network Function Virtualization (NFV) and the OSI Architecture
3. Artificial Intelligence for Network Traffic Management
4. Machine Learning-Based Error Detection in Networks
5. Digital Twin Networks and the OSI Model
6. 6G Communication Architecture Compared with the OSI Model
7. Edge Computing and Reliable Data Transmission
8. Cloud Networking Through the OSI Architecture
9. Satellite Internet Communication Using OSI Layers
10. Future of Reliable Networking Beyond the Traditional OSI Model

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable	Basic analysis with limited insights	Poor or incorrect analysis of results

			interpretation		
Documentation	10	Clear, well structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation